

In []:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),
        'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

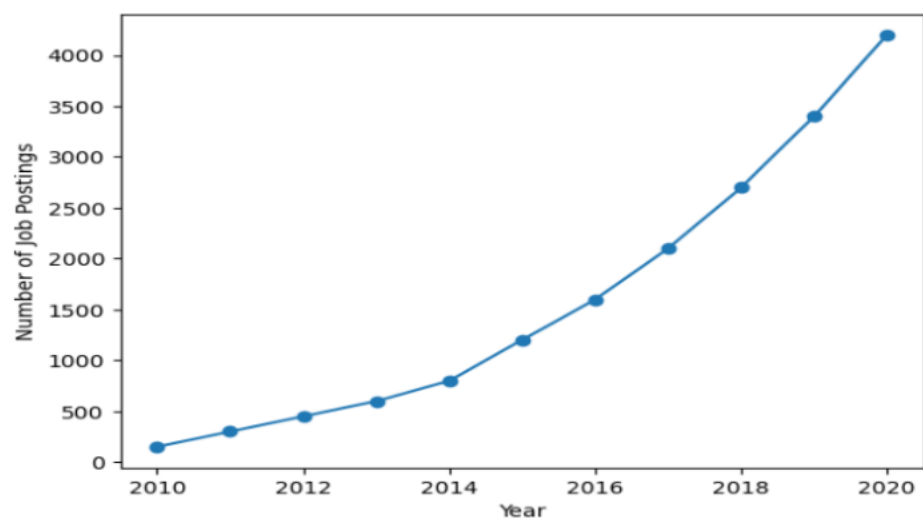
plt.title("Trend of Data Science Job Postings")

plt.xlabel('Year')

plt.ylabel('Number of Job Postings')

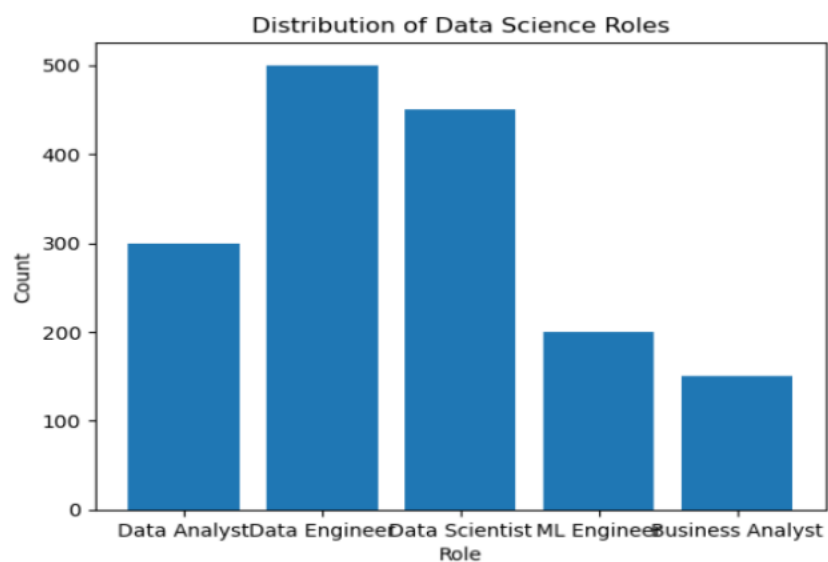
plt.show()
```

```
In[]
```



In []:

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',  
        'Business Analyst']  
counts = [300, 500, 450, 200, 150]  
plt.bar(roles, counts)  
plt.title('Distribution of Data Science Roles')  
plt.xlabel('Role')  
plt.ylabel('Count')  
plt.show()
```



In []:

```
# Structured data example
structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,
or a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Charlie	35

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

```
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
}
```

In []:

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhylZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAAn
XEOoSzaNRINI4zAbNK8lgx2pprrNn-j6agbWdf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kvv8GYDSI8FovltV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```